

Aapo Rotonen

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SUMMARY

Test Automation & Software Engineer with 4 years of experience in the development and validation of **commercial safety-critical software**. Specialist in C++ feature development and architecting **hardware-integrated Release Test Automation** frameworks using Python to ensure **ISO 26262** compliance. Experienced in AI-driven workflows, currently managing a global digital media project with 250M+ total views by architecting automated content pipelines (LLMs, Voice, and Video). Currently expanding expertise in Cloud Infrastructure and AI-enhanced quality assurance.

TECHNICAL SKILLS

- **Software Development:** C++, Python, Qt/QML, JavaScript, TypeScript, Node.js, React, Shell scripting.
- **Quality Assurance & Automation:** Robot Framework, Selenium, Playwright, Squish, API Testing (Postman), Jenkins (CI/CD). **Methodologies:** White-box & Black-box testing, Manual & Exploratory testing, Edge cases & Boundary values, Regression & Unit testing.
- **AI & Data Engineering:** Generative AI Pipelines (LLMs, Voice, Video), AI Workflow Design, SQL (PostgreSQL), Data Analytics & Platform Algorithms.
- **Standards & Infrastructure:** ISO 26262, MISRA C++ 2023, Azure (AZ-900/AI-900 in progress), Containerization (Docker), Linux, QNX, Git, Gerrit, Jira.

WORK EXPERIENCE

Qt Group | Software Engineer | Sep 2022 – Nov 2025

- Developed features, demos, and fixed bugs for a safety-critical project using C++, Qt, and QML.
- Designed and implemented a hardware-integrated test automation architecture (**Python & Squish**) that streamlined the release process.
- Managed the full validation cycle: test plan design (**TestRail**), code reviews, and automated execution in **Jenkins**.
- Collaborated with cross-functional teams, including Product Management and R&D, to align feature development with technical requirements and safety standards.
- Ensured full traceability by drafting documentations and test reports in accordance with safety-critical documentation standards.

Qt Group | Trainee & Thesis Worker | Nov 2021 – Sep 2022

- Designed and implemented a hardware-integrated Release Test Automation solution as part of a Bachelor's Thesis, laying the foundation for automated system-level validation.
- Supported development through scripting, debugging, bug fixing, and developing features and demos.

PROJECTS (AI & AUTOMATION)

Founder & AI Content Strategist | AI-Driven Digital Media Project

- **Scale:** Developed and scaled a global media project to **250M+ total views** (120M monthly views at peak) on YouTube & TikTok.
- **AI Workflow Design:** Designed a high-efficiency production pipeline by leveraging advanced Generative AI tools (e.g. ChatGPT, Gemini, OpenArt, Pollo.ai, Hailuo AI, Grok, ElevenLabs, and more) and their AI models.
- **Tool Orchestration:** Managed multiple AI models for visual consistency, script generation, and high-fidelity audio to maintain a daily production cycle.
- **Data-Driven Growth:** Analyzed platform metrics to optimize viewer retention and CTR.

PROJECTS (Full-Stack)

E-commerce Platform

- Developed a full-stack marketplace using **React and Node.js**. Designed and implemented **RESTful APIs** and used **HTML/CSS** to create a secure, dynamic user experience with robust backend logic.

Interactive Portfolio

- Developed a responsive website using a **modern JavaScript-based stack (React, HTML, CSS)** to showcase technical skills and professional history.

EDUCATION & CERTIFICATIONS

- **Bachelor of Engineering, IT** | Oulu University of Applied Sciences (2019–2022)
- **Microsoft Certified:** Azure Fundamentals (AZ-900) – *In progress*
- **Microsoft Certified:** Azure AI Fundamentals (AI-900) – *In progress*
- **ISTQB Foundation Level** – *Self-study*
- **Languages:** Finnish (Native), English (Full Professional Proficiency)